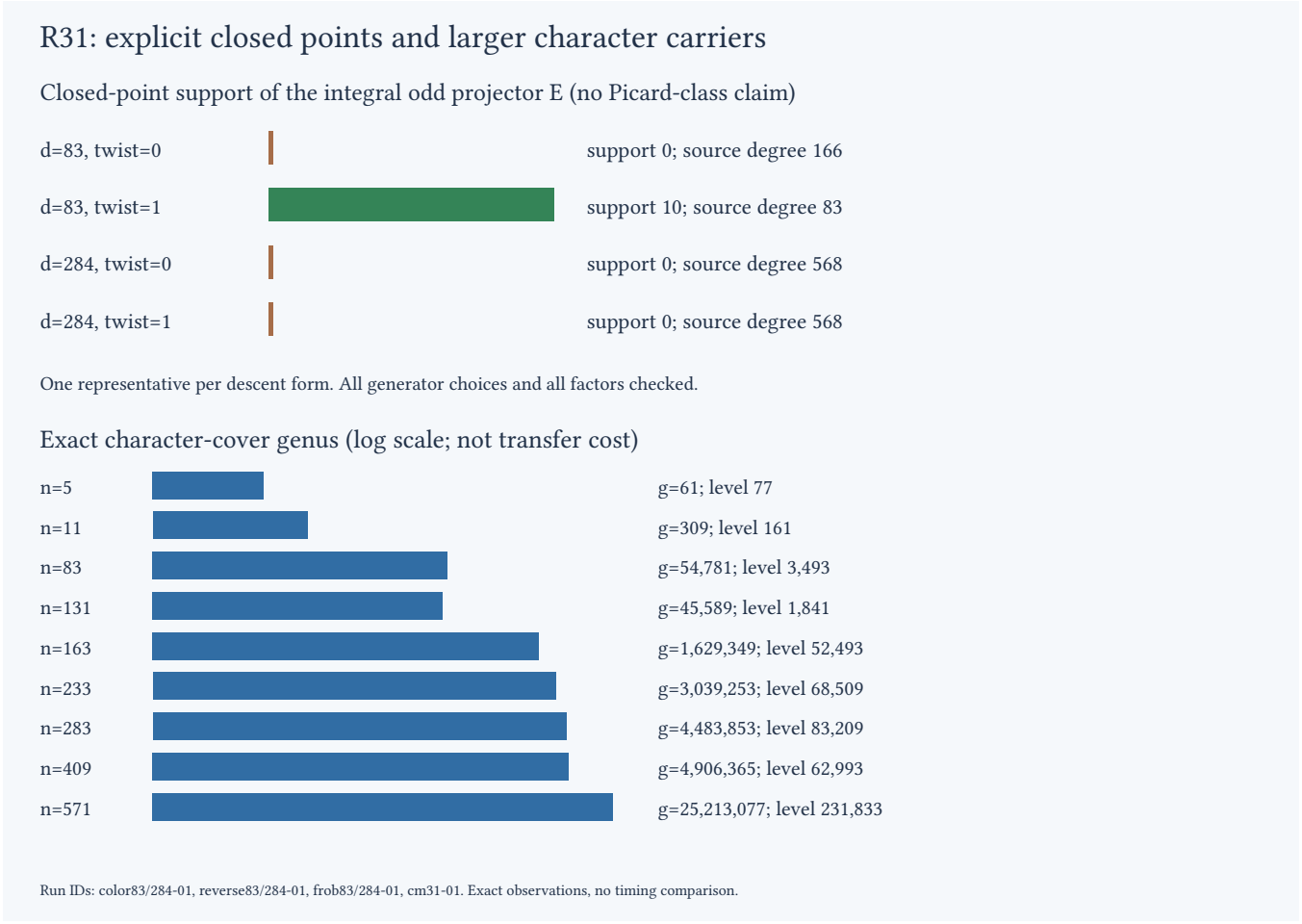
Koblitz cover transfers — R31

Exact observations and structural proofs. The research goal continues.

Two remaining ordinary inputs have completely enumerated middles and full 49-level transport checks. Their kernel colors are (3,0) at degree83 and (4,0) at degree284. The zero second color supplies five rational generator abscissas, which are now passed into the original Tate/QR incidence equations.



Every generator and both descent forms satisfy all five original residuals, with all eight registered denominators units, exact order11, ten diamond iterates, the fifth-iterate sign, and corrupted-nu rejection. These are actual finite-algebra point certificates. They are not a global map or an evaluated nonzero Jacobian image.

A surviving divisor candidate and explicit limits

On the original degree83 lane the source points have degree166. The odd projector $E=(1-T^5)(5\text{-sum } T^{2i})$ cancels on closed points. On the diamond twist the source points have degree83, ten distinct closed diamond translates, and nonzero signed formal weights. Both degree284 descent forms have degree568 and cancel. Every irreducible factor of every orientation quadratic was included. Generator labels and orientation choices give repeated candidates, not independent divisor classes.

The new task is proper divisor-class evaluation on the degree83 diamond twist. Nonzero formal weights do not prove nonprincipality; nonprincipality alone would not identify the intended order11 modular factor. Retain the actual coordinates, Frobenius classes and signed weights through that test.

Larger character data

The existing one-prime CM construction is extended to n163,233,283,409,571, with n5,11,83,131 as controls. Full conjugate-root-label bijections certify the recorded Euler factors; n5/n11 also have dense ring norm controls. Cyclic fiber cocycles provide exact genus calculations without enlarging the old ten-million-state materialization cap. Complete materialization is checked on the smaller controls. Connectedness has retained Bezout data.

The base permutation arrays were checked internally but not serialized as promised prospectively. Source, cycle histograms and Bezout data remain; this is a serialization gap and no independent replay certificate is claimed. Explicit larger curve equations and subgroup transfer remain open.

Proof and evidence scope

Ten named Lean structural lemmas cover cyclic translation order and periods, product relabelling, rational Tate identities, QR coefficient changes under the diamond twist and the sign involution. Compile/audit details are in lean31-01. These do not formally verify the native finite-field tables, modularity, normalization, genus geometry or Picard nonprincipality.

Seven native attempts are accepted. All mathematical outputs are native Rust/Singular; Python only assembles text, indexes metadata and renders. No old failed source or archive is changed. No EKS/Diem/freeze restart, new coordination claim, merge or auto-merge occurred. Root-library validation is not newly certified; the earlier inherited test limitations still apply.

References and precise prospective contracts: REPORT.md, PROTOCOL.md, REVERSE31-PROTOCOL.md, FROB31-PROTOCOL.md, PROOF_MAP.md and NEXT.md. No IC timing, rank, key recovery or attack conclusion is made.